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Recursive Subdivision & Measures on Projective Line

*Invariant Measures Induced by Harmonic and Mediant Trees
on the Projective Line of Slopes*

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Companion diagram: leelemacjames.github.io/ergodic-projective.html

Abstract. Bertrand's paradox shows that geometry alone does not determine probability: different sampling rules on the same circle give different chord-length probabilities, and the classical answers $1/2$, $1/3$, $1/4$ arise from genuinely different parametrisations of the two-dimensional space of chords. If we fix one endpoint of the chord at a boundary point $P = (-1, 0)$, the paradox dissolves: the space of chord directions becomes a one-dimensional projective line of slopes, and the natural Euclidean uniform measure on this pencil is essentially unique (the uniform angular measure $d\theta$). The lesson of Bertrand nevertheless persists. Even on this simpler one-dimensional sample space, there are further natural probability measures that arise not from Euclidean sampling but from purely arithmetic recursive subdivision of the slope axis. The Farey mediant rule $(a/b, c/d) \mapsto (a+c)/(b+d)$ generates the Stern–Brocot tree, whose breadth-first traversal induces the classical Minkowski question-mark measure: continuous, strictly increasing, but singular with respect to Lebesgue measure, with derivative zero almost everywhere. The harmonic mean rule $(a/b, c/d) \mapsto 2ac/(ad+bc)$ generates the harmonic interpolation tree, whose breadth-first traversal induces an absolutely continuous measure with density $\rho(t) \propto 1/t^2$. The two trees produce Pythagorean triples on the unit circle via the line-from- $(-1, 0)$ construction, and the harmonic tree is shown to be the unique projectively natural midpoint operation, characterised by the cross-ratio condition $(t_1, t_2; 0, t) = -1$. Along its central path from seeds $1/2$ and $2/3$, the harmonic tree generates a Mersenne-indexed family of primitive triples with slopes $2^{n+1}/M_{n+2}$ where $M_k = 2^k - 1$. The headline contribution is the comparison: even where Euclidean geometry offers only one natural measure, recursive subdivision rules *themselves* induce further probability measures on $\mathbb{P}^1(\mathbb{R})$, and the choice of rule determines whether the limit is singular (Farey) or absolutely continuous (harmonic).

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1. Bertrand reformulated: measures on the projective line of chords

Bertrand’s chord paradox asks: what is the probability that a random chord of the unit circle is longer than the side of the inscribed $1/3$, or $1/4$ depending on how chords are sampled, demonstrate that the question is not well posed until a probability measure on the space of chords is specified. The paradox is not about geometry but about the choice of measure on a geometric space.

The simplest version of this lesson restricts attention to chords through a fixed boundary point. Fix $P = (-1, 0)$ on the unit circle. Every direction from P determines a unique chord. The space of directions is the projective line $\mathbb{P}^1(\mathbb{R})$, parametrised by the slope t of the line from P . A probability measure on $\mathbb{P}^1(\mathbb{R})$ then induces a probability distribution on chords through P .

The standard Bertrand sampling rules apply to the full two-dimensional space of chords, where the multiplicity of natural parametrisations gives the paradox its force. Restricting to the one-dimensional pencil of chords through a fixed boundary point collapses much of that multiplicity: on this pencil, the natural Euclidean uniform measure is essentially unique. If a chord is sampled by drawing its second endpoint uniformly at random from the circumference, the induced measure on slopes is the uniform angular measure $d\mu_\theta \propto d\theta$, which in slope coordinates t is the Cauchy measure $d\mu_\theta \propto dt/(1 + t^2)$. The other classical Bertrand procedures (uniform midpoint over the disc, uniform radial coordinate along a diameter) do not naturally restrict to a one-dimensional pencil through a fixed point: they sample the two-dimensional chord space and have no canonical interpretation when the basepoint is fixed.

Fixing the basepoint, then, simplifies the original paradox: the multiplicity of natural Euclidean measures is reduced to one. The lesson of Bertrand, however, persists in a

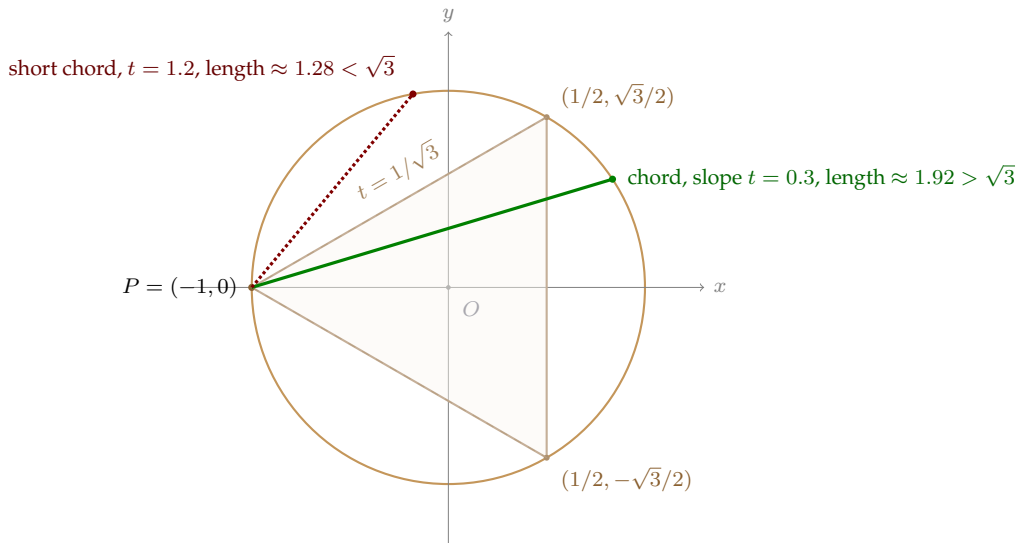


Figure 1: The Bertrand setup with a fixed boundary point. The unit circle, the inscribed equilateral triangle with one vertex at the basepoint $P = (-1, 0)$, and two sample chords from P . The dashed lines to the two upper-and-lower triangle vertices correspond to slopes $t = \pm 1/\sqrt{3}$; chords with $|t| < 1/\sqrt{3}$ (green, solid) exit on the far arc and are longer than the triangle side $\sqrt{3}$; chords with $|t| > 1/\sqrt{3}$ (red, dotted) exit on the near arc and are shorter. The Bertrand question is whether the probability $\Pr(|t| < 1/\sqrt{3})$ is well defined; the answer depends entirely on which probability measure is placed on the slope axis $\mathbb{P}^1(\mathbb{R})$.

different form. Even on this one-dimensional pencil, where Euclidean geometry singles out only the uniform angular measure, other natural probability measures exist, arising not from any continuous Euclidean procedure but from purely arithmetic recursive subdivision of the slope axis itself. Two such operations present themselves:

- The **Farey mediant**: $(a/b, c/d) \mapsto (a + c)/(b + d)$.
- The **harmonic mean**: $(a/b, c/d) \mapsto 2ac/(ad + bc)$.

Each operation takes two rational slopes and produces a third lying strictly between them. Iterated, each generates a binary tree of rationals dense in any seed interval. The breadth-first traversal of each tree induces an empirical probability measure on the seed interval, and the limit of these empirical measures as depth grows is the natural invariant measure of the recursion.

These two recursive measures are the subject of this paper. The Farey mediant is the classical operation underlying the Stern–Brocot tree and the enumeration of all positive rationals; the harmonic mean is a projectively natural alternative whose tree is less familiar but whose induced measure has a much simpler analytic form.

Once the classical ambiguity is removed by fixing a boundary point, the question is no longer which Euclidean measure to choose, but which recursive subdivision should define the measure. In this setting the Farey mediant and harmonic interpolant induce fundamentally different invariant measures on the same projective space. The comparison is the main result.

Remark 1.1 (Why fixing the basepoint dissolves the original Bertrand multiplicity). The classical Bertrand paradox draws its force from the two-dimensional space of chords admitting multiple genuinely inequivalent natural parametrisations: endpoints on the circumference, midpoints in the disc, radial coordinates along a fixed diameter. Each parametrisation has its own natural uniform measure, and the three uniform measures push forward to genuinely different distributions on chord length. The paradox is that there is no intrinsic reason to prefer one parametrisation. When we restrict to chords through a fixed point P , the chord-space dimensionality collapses from two to one, and the natural Euclidean parametrisation becomes the single circumference-angle of the other endpoint. The multiplicity of classical Bertrand answers does not survive the restriction. What this paper observes is that even on this simpler one-dimensional space, recursive arithmetic subdivision provides further natural measures whose regularity classes are genuinely different from the uniform Euclidean one and from each other.

2. Slopes, intersections, and Pythagorean triples

The unit circle $x^2 + y^2 = 1$ and the boundary point $(-1, 0)$ are the geometric setting. A line through $(-1, 0)$ with slope t has equation $y = t(x + 1)$, and substituting into the circle equation gives

$$(1 + t^2)x^2 + 2t^2x + (t^2 - 1) = 0,$$

with roots $x = -1$ and $x = (1 - t^2)/(1 + t^2)$, yielding the second intersection

$$\left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right). \quad (1)$$

The slope t is the rise-over-run of the chord. Three special slopes have direct geometric meaning. Slope 0 is the horizontal chord through $(-1, 0)$ to the antipode $(1, 0)$. Slope ∞ is the vertical tangent line at $(-1, 0)$, which we treat as the projective limit returning to the starting point. Slope 1 is the chord rising at 45° to the top of the circle at $(0, 1)$. As t ranges over $\mathbb{R} \cup \{\infty\}$, the second intersection traces out the circle exactly once. The slope axis $\mathbb{P}^1(\mathbb{R}) = \mathbb{R} \cup \{\infty\}$ is therefore a complete coordinate for the circle relative to the basepoint $(-1, 0)$, and the chord-direction space of the previous section is concretely this slope axis.

2.1 Pythagorean triples from rational slopes

For a rational slope $t = p/q$ with $\gcd(|p|, q) = 1$ and $q > 0$, clearing denominators in (1) gives the homogeneous coordinates

$$[p^2 - q^2 : 2pq : p^2 + q^2] \in \mathbb{P}^2(\mathbb{R}). \quad (2)$$

A direct check shows $(p^2 - q^2)^2 + (2pq)^2 = (p^2 + q^2)^2$, so this is a primitive Pythagorean triple. Every primitive triple arises from a unique rational slope, making the map $t \mapsto [\cdot]$ a bijection between $\mathbb{Q}$ and the primitive triples on the conic $a^2 + b^2 = c^2$. Slope $1/2$ gives $[3 : 4 : 5]$, slope $2/3$ gives $[5 : 12 : 13]$, slope $3/4$ gives $[7 : 24 : 25]$, slope $4/7$ gives $[33 : 56 : 65]$. The Bertrand question on chord directions becomes, in the rational case, a question about which Pythagorean triples are generated by which subdivision rule.

2.2 The Farey mediant in concrete terms

The first recursive subdivision rule of this paper is the Farey mediant. Given two slopes p_1/q_1 and p_2/q_2 with $p_1/q_1 < p_2/q_2$, the mediant is obtained by adding numerators and

denominators separately:

$$\frac{p_1}{q_1} \oplus \frac{p_2}{q_2} = \frac{p_1 + p_2}{q_1 + q_2}. \quad (3)$$

The mediant lies strictly between the two inputs on the real line. For seeds $1/2$ and $2/3$, the mediant is $3/5$. Iterating the mediant operation between $0/1$ and $1/0$ (or between any two Farey-neighbour seeds) generates the Stern–Brocot tree, which enumerates every positive rational in lowest terms exactly once.

The mediant is a clean arithmetic operation but a coordinate-dependent one. The mediant of $1/2$ and $2/3$ is $3/5$; the mediant of the equivalent fractions $2/4$ and $4/6$ is $6/10$, which simplifies back to $3/5$, but the mediant of $1/2$ and $4/6$ is $5/8$, a different rational. The output depends not only on the slope values but on which integer representatives are chosen. From a projective viewpoint, where a slope is a value in $\mathbb{P}^1(\mathbb{R})$ and not a chosen pair of integers, the mediant is not an intrinsic operation on slopes.

This is the strength of the Stern–Brocot tree in that it exploits the integer-lattice structure. But it motivates the question: is there a subdivision rule that depends only on the slope values and not on their integer representatives? The harmonic mean is one answer.

3. The harmonic mean as projective midpoint

The harmonic mean of two positive slopes t_1 and t_2 is

$$H(t_1, t_2) = \frac{2t_1 t_2}{t_1 + t_2} = \frac{2pr}{ps + qr}, \quad t_1 = \frac{p}{q}, \quad t_2 = \frac{r}{s}. \quad (4)$$

It lies strictly between t_1 and t_2 whenever the two are distinct positive numbers. The harmonic mean of $1/2$ and $2/3$ is $4/7$; of $1/2$ and $1/3$ is $2/5$; of $3/5$ and $2/3$ is $12/19$. The value of H depends only on the slope values, not on the choice of integer representatives: scaling either input by a common factor leaves H unchanged, because the factor cancels between numerator and denominator in (4).

3.1 Projective basepoints and cross-ratio

The slope axis $\mathbb{R} \cup \{\infty\}$ has two distinguished values arising directly from the circle construction: slope 0 is the horizontal chord from $(-1, 0)$ to the antipode $(1, 0)$, and slope ∞ is the vertical tangent line corresponding back to the basepoint $(-1, 0)$. These two slopes pick out the extremal directions of the construction, the chord and the tangent at the basepoint.

The projective line $\mathbb{P}^1(\mathbb{R})$ has a canonical four-point invariant, the cross-ratio

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

The cross-ratio is preserved by every Möbius transformation $z \mapsto (az + b)/(cz + d)$, and a cross-ratio of -1 singles out a harmonic range: z_4 is the harmonic conjugate of z_3 with respect to (z_1, z_2) .

Theorem 3.1 (Cross-ratio characterisation of the harmonic mean). *For positive slopes t_1 and t_2 , the harmonic mean $H(t_1, t_2)$ is the unique slope $t \in \mathbb{P}^1(\mathbb{R})$ for which*

$$(t_1, t_2; 0, t) = -1.$$

Equivalently, $H(t_1, t_2)$ is the harmonic conjugate of the horizontal-chord slope 0 with respect to the pair (t_1, t_2) .

Proof. Setting $(t_1, t_2; 0, t) = -1$:

$$\frac{(t_1 - 0)(t_2 - t)}{(t_1 - t)(t_2 - 0)} = -1 \implies t_1(t_2 - t) = -t_2(t_1 - t),$$

which expands to $t_1 t_2 - t_1 t = -t_1 t_2 + t_2 t$, giving $2t_1 t_2 = t(t_1 + t_2)$, hence $t = 2t_1 t_2 / (t_1 + t_2) = H(t_1, t_2)$. Uniqueness follows because the cross-ratio is a Möbius transformation in the fourth argument when the other three are fixed, hence a bijection on $\mathbb{P}^1(\mathbb{R})$. $\square$

The harmonic mean is a projectively natural midpoint operation on slopes relative to the geometric basepoints 0 (the horizontal chord to the antipode) and ∞ (the tangent at the starting point). The Farey mediant has no such characterisation.

Remark 3.2 (Elementary form). The harmonic mean also satisfies the reciprocal identity

$$\frac{1}{H(t_1, t_2)} = \frac{1}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right),$$

the average of the reciprocals. This is the form familiar from elementary physics, where it gives the effective resistance of two resistors in parallel and the mean speed of a round trip at two different speeds. The cross-ratio characterisation in Theorem 3.1 elevates this familiar operation to a projectively intrinsic role on the slope axis.

3.2 The harmonic tree of slopes

Given seed slopes $t_1 < t_2$, the harmonic tree is constructed by repeated insertion of H . At depth d the tree contains 2^{d-1} new slopes; total nodes to depth d amount to $2^d - 1$. The harmonic insertion divides $[t_1, t_2]$ asymmetrically:

$$H(t_1, t_2) - t_1 = \frac{t_1(t_2 - t_1)}{t_1 + t_2}, \quad t_2 - H(t_1, t_2) = \frac{t_2(t_2 - t_1)}{t_1 + t_2},$$

giving distance ratio $t_1 : t_2$. The smaller-slope subinterval is shorter than the larger-slope subinterval by exactly the factor t_1/t_2 . For seeds $1/2$ and $2/3$, the insertion $4/7$ lies a little closer to $1/2$ than to $2/3$, in ratio $3 : 4$. This asymmetric splitting propagates to every level of the tree and is the key structural feature governing the induced measure to be explored in section 5.

4. Central paths in the two trees: arithmetic versus geometric growth

Before turning to the induced measures, we examine a concrete structural feature of each tree: the path of slopes generated by always inserting against the same fixed seed. This central path is the simplest possible traversal of each tree, and the two trees give denominator sequences of fundamentally different growth rates.

4.1 The Stern–Brocot central path: arithmetic progression

Theorem 4.1 (Linear denominator growth on the Stern–Brocot central path). Define the central path of the Stern–Brocot tree from seeds $t_1 = 1/2$ and $t_2 = 2/3$ by setting $s_0 = 2/3$ and

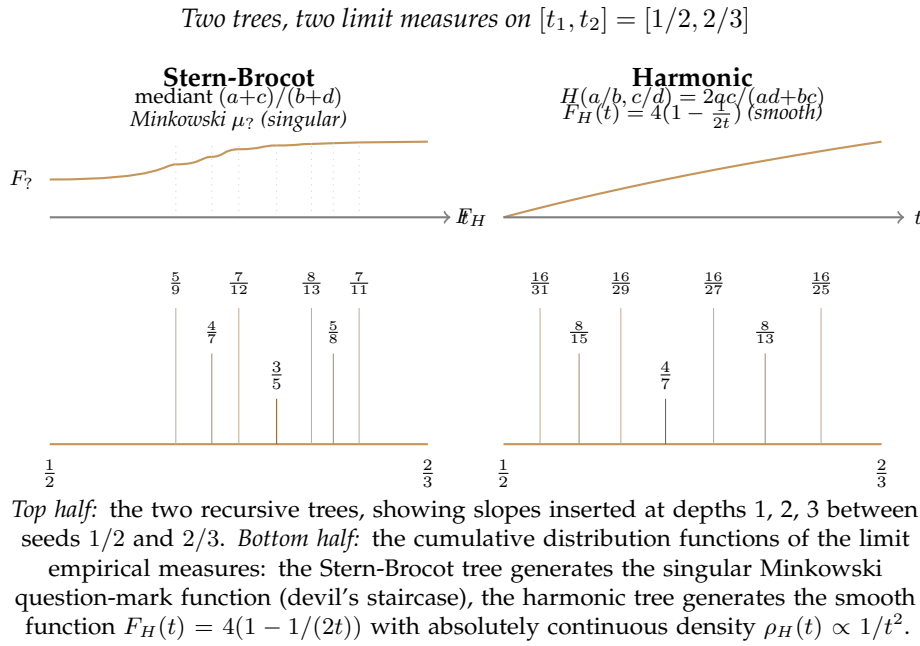


Figure 2: Tree subdivisions and their limit measures (see section 5)

$s_{n+1} = \text{mediant}(1/2, s_n)$. Then for $n \geq 0$,

$$s_n = \frac{n+2}{2n+3}.$$

The corresponding primitive Pythagorean triples are $[|(n+2)^2 - (2n+3)^2| : 2(n+2)(2n+3) : (n+2)^2 + (2n+3)^2]$ reduced.

Proof. By induction. At $n = 0$, $s_0 = 2/3 = (0+2)/(0 \cdot 2 + 3)$, matching. Assume $s_n = (n+2)/(2n+3)$. Then

$$s_{n+1} = \text{mediant}(1/2, s_n) = \frac{1 + (n+2)}{2 + (2n+3)} = \frac{n+3}{2n+5},$$

matching the formula at $n+1$ with $n+3 = (n+1) + 2$ and $2n+5 = 2(n+1) + 3$. $\square$

The denominators along the Stern–Brocot central path from $1/2$ are 3, 5, 7, 9, 11, 13, 15, $\dots$, an arithmetic progression with common difference 2. The Pythagorean triples grow polynomially in n : $[5 : 12 : 13]$, $[8 : 15 : 17]$, $[33 : 56 : 65]$, $[28 : 45 : 53]$, $[85 : 132 : 157]$, $[60 : 91 : 109]$, $[161 : 240 : 289]$, $\dots$. The triples include neither a clean indexed family nor a recognisable arithmetic structure.

4.2 The harmonic central path: Mersenne denominators

Theorem 4.2 (Mersenne denominator growth on the harmonic central path). Define the central path of the harmonic tree from seeds $t_1 = 1/2$ and $t_2 = 2/3$ by setting $h_0 = 2/3$ and $h_{n+1} = H(1/2, h_n)$. Then for $n \geq 0$,

$$h_n = \frac{2^{n+1}}{2^{n+2} - 1} = \frac{2^{n+1}}{M_{n+2}},$$

where $M_k = 2^k - 1$ is the k -th Mersenne number. The corresponding primitive Pythagorean triple is

$$[a_n : b_n : c_n] = [M_{n+2}^2 - 4^{n+1} : 2^{n+2}M_{n+2} : M_{n+2}^2 + 4^{n+1}].$$

Proof. By induction. At $n = 0$, $h_0 = 2/3 = 2^1/(2^2 - 1)$, matching. Suppose $h_n = 2^{n+1}/(2^{n+2} - 1)$. The central insertion is

$$h_{n+1} = H(1/2, h_n) = \frac{2 \cdot \frac{1}{2} \cdot h_n}{\frac{1}{2} + h_n} = \frac{h_n}{\frac{1}{2} + h_n}.$$

Substituting:

$$h_{n+1} = \frac{2^{n+1}/(2^{n+2} - 1)}{\frac{1}{2} + 2^{n+1}/(2^{n+2} - 1)} = \frac{2^{n+1}}{\frac{1}{2}(2^{n+2} - 1) + 2^{n+1}} = \frac{2^{n+2}}{2^{n+3} - 1},$$

matching the formula at $n + 1$. The triple formula follows from (2) with $p = 2^{n+1}$ and $q = M_{n+2}$. $\square$

n	SB slope s_n	SB triple	Harmonic slope h_n	Harmonic triple
0	2/3	[5 : 12 : 13]	2/3	[5 : 12 : 13]
1	3/5	[8 : 15 : 17]	4/7	[33 : 56 : 65]
2	4/7	[33 : 56 : 65]	8/15	[161 : 240 : 289]
3	5/9	[28 : 45 : 53]	16/31	[705 : 992 : 1217]
4	6/11	[85 : 132 : 157]	32/63	[2945 : 4032 : 4993]
5	7/13	[60 : 91 : 109]	64/127	[12033 : 16256 : 20225]
6	8/15	[161 : 240 : 289]	128/255	[48641 : 65280 : 81409]

The contrast is stark. The Stern–Brocot central path advances by a linear increment in both numerator and denominator, generating triples with hypotenuses growing polynomially as $O(n^2)$. The harmonic central path advances by doubling the numerator and approximately doubling the denominator, generating triples with hypotenuses growing exponentially as $O(4^n)$. Despite their shared starting slopes the two paths diverge immediately and visit entirely different families of triples thereafter.

Remark 4.3 (The Fibonacci structure of Stern–Brocot lives elsewhere). The famous Fibonacci-denominator sequence $1/2, 2/3, 3/5, 5/8, 8/13, 13/21, \dots$ of Stern–Brocot is *not* the central path from a fixed seed. It is the *zigzag path*: starting at root $1/1$ and alternating left and right children at each level, the path passes through consecutive Fibonacci ratios F_n/F_{n+1} and converges to the golden ratio conjugate $1/\varphi = (\sqrt{5} - 1)/2$. This is the path of best rational approximants to $1/\varphi$ in the Stern–Brocot tree, and reflects the fact that the continued fraction expansion of φ is $[1; 1, 1, 1, \dots]$. The Fibonacci structure of Stern–Brocot captures approximation to a specific quadratic irrational; it is not the structure of the central path from an arbitrary seed.

Remark 4.4 (Mersenne primes on the harmonic central path). A Mersenne number $M_k = 2^k - 1$ is prime only for specific k : $k = 2, 3, 5, 7, 13, 17, 19, 31, \dots$. At the depths where M_{n+2} is prime (namely $n = 0, 1, 3, 5, 11, 15, 17, 29, \dots$), the denominator of the harmonic central-path slope is a Mersenne prime, and the triple has unusually clean arithmetic. At $n = 1$, $M_3 = 7$ is prime, giving $[33 : 56 : 65]$ with $65 = 7^2 + 4^2$. At $n = 3$,

$M_5 = 31$ is prime, giving $[705 : 992 : 1217]$ with $1217 = 31^2 + 4^3$. At $n = 5$, $M_7 = 127$ is prime, giving $[12033 : 16256 : 20225]$ with $20225 = 127^2 + 4^6$. The harmonic central path picks out a Mersenne-indexed subsequence of primitive Pythagorean triples without any extrinsic appeal to Mersenne primes; the structure is built into the recursion itself.

5. The measures induced by the two trees

We now arrive at the central comparison. Each recursive subdivision tree, traversed breadth-first, produces a sequence of rational slopes dense in the seed interval. The empirical probability measure of this sequence converges, as the depth grows, to a well-defined invariant measure on the interval. The two trees give measures of strikingly different analytic character.

5.1 The harmonic tree: absolutely continuous $1/t^2$ density

Theorem 5.1 (Empirical measure of the harmonic tree). *The empirical distribution of the harmonic tree between seeds $t_1 < t_2$ converges weakly, as the depth grows, to the probability measure on $[t_1, t_2]$ with density*

$$\rho_H(t) = \frac{t_1 t_2}{t_2 - t_1} \cdot \frac{1}{t^2}.$$

This measure is absolutely continuous with respect to Lebesgue measure, with cumulative distribution function

$$F_H(s) = \frac{t_2}{t_2 - t_1} \left(1 - \frac{t_1}{s}\right), \quad s \in [t_1, t_2].$$

The proof proceeds in three steps: first establish the self-similarity of the limiting measure under the recursive splitting, then translate the self-similarity into a local differential condition on the density, then solve the resulting first-order equation.

Proof. Step 1: self-similarity of the empirical measure. At depth 1 the tree inserts $H(t_1, t_2)$, dividing the interval $[t_1, t_2]$ into two subintervals of lengths $\ell_1 = t_1(t_2 - t_1)/(t_1 + t_2)$ and $\ell_2 = t_2(t_2 - t_1)/(t_1 + t_2)$, with $\ell_1 + \ell_2 = t_2 - t_1$ and $\ell_1/\ell_2 = t_1/t_2$. Note that $\ell_1 \neq \ell_2$ unless $t_1 = t_2$: the harmonic insertion is never at the geometric midpoint of the interval.

The breadth-first enumeration visits all nodes at depth 1 before any at depth 2, all at depth 2 before any at depth 3, and so on. The fraction of nodes lying in $[t_1, H(t_1, t_2)]$ at depths 1 through d tends to $1/2$ as $d \rightarrow \infty$, since each subinterval is itself filled by a binary subtree with $2^{d-1} - 1$ internal nodes at depth d . The same holds for the right subinterval. The limiting probability measure μ_H therefore satisfies

$$\mu_H([t_1, H(t_1, t_2)]) = \mu_H([H(t_1, t_2), t_2]) = \frac{1}{2}. \quad (5)$$

This self-similarity applies recursively to each subinterval: the same binary tree construction within $[t_1, H]$ produces a measure on that subinterval satisfying its own version of (5), and similarly for $[H, t_2]$.

Step 2: local differential condition on the density. Assume μ_H has a continuous positive density $\rho(t)$ on (t_1, t_2) . Apply (5) to an arbitrary subinterval $[a, b]$ generated by the tree, $t_1 \leq a < b \leq t_2$, writing $H_{ab} = H(a, b) = 2ab/(a + b)$:

$$\int_a^{H_{ab}} \rho(t) dt = \int_{H_{ab}}^b \rho(t) dt.$$

Set $b = a + \epsilon$ for small $\epsilon > 0$. Taylor-expanding $H_{a,a+\epsilon} = 2a(a + \epsilon)/(2a + \epsilon)$:

$$\begin{aligned} H_{a,a+\epsilon} &= \frac{2a^2 + 2a\epsilon}{2a + \epsilon} = a + \frac{a\epsilon}{2a + \epsilon} = a + \frac{\epsilon}{2} \left(1 - \frac{\epsilon}{2a} + O(\epsilon^2)\right) \\ &= a + \frac{\epsilon}{2} - \frac{\epsilon^2}{4a} + O(\epsilon^3). \end{aligned}$$

The left subinterval $[a, H_{a,a+\epsilon}]$ has length $\epsilon/2 - \epsilon^2/(4a) + O(\epsilon^3)$, and the right subinterval $[H_{a,a+\epsilon}, a + \epsilon]$ has length $\epsilon/2 + \epsilon^2/(4a) + O(\epsilon^3)$. Approximating the density by its value at a on each subinterval (with corrections in $\rho(a + \epsilon) = \rho(a) + \epsilon\rho'(a) + O(\epsilon^2)$), the equal-mass condition becomes:

$$\rho(a) \left(\frac{\epsilon}{2} - \frac{\epsilon^2}{4a} \right) = (\rho(a) + \epsilon\rho'(a)) \left(\frac{\epsilon}{2} + \frac{\epsilon^2}{4a} \right) + O(\epsilon^3).$$

The ϵ^1 terms cancel: $\rho(a) \cdot \epsilon/2 = \rho(a) \cdot \epsilon/2$. At order ϵ^2 :

$$-\frac{\rho(a)}{4a} = \frac{\rho(a)}{4a} + \frac{\rho'(a)}{2},$$

which rearranges to

$$\frac{\rho'(a)}{\rho(a)} = -\frac{1}{a}. \quad (6)$$

Step 3: solve the differential equation. Integrating $\rho'/\rho = -1/t$ gives $\log \rho(t) = -\log t + C$, so

$$\rho(t) = \frac{K}{t^2}$$

for some constant $K > 0$. Normalisation $\int_{t_1}^{t_2} \rho(t) dt = 1$ gives

$$K \int_{t_1}^{t_2} \frac{dt}{t^2} = K \left(\frac{1}{t_1} - \frac{1}{t_2} \right) = K \cdot \frac{t_2 - t_1}{t_1 t_2} = 1,$$

so $K = t_1 t_2 / (t_2 - t_1)$, giving the density in the theorem. Integrating from t_1 to s yields the cumulative distribution function:

$$F_H(s) = \frac{t_1 t_2}{t_2 - t_1} \int_{t_1}^s \frac{dt}{t^2} = \frac{t_1 t_2}{t_2 - t_1} \cdot \left(\frac{1}{t_1} - \frac{1}{s} \right) = \frac{t_2}{t_2 - t_1} \left(1 - \frac{t_1}{s} \right). \quad \square$$

□

5.2 Numerical verification

Generating all $2^{12} - 1 = 4095$ nodes of the harmonic tree between seeds $t_1 = 1/2$ and $t_2 = 2/3$ to depth 12 and comparing the empirical cumulative distribution function with the theoretical formula $F_H(s) = 4(1 - 1/(2s))$ (using $t_2/(t_2 - t_1) = (2/3)/(1/6) = 4$) gives close agreement at every test point in the interval. The same construction applied to the Stern–Brocot tree between the same seeds produces a strikingly different empirical CDF, anticipating the singular measure described in Section 5.3.

s	Empirical $F_H(s)$	Theoretical $F_H(s)$	Difference	Empirical $F_?(s)$
0.5100	0.078388	0.078431	4.3×10^{-5}	0.000000
0.5300	0.226374	0.226415	4.1×10^{-5}	0.008059
0.5500	0.363614	0.363636	2.2×10^{-5}	0.093773
0.5700	0.491331	0.491228	1.0×10^{-4}	0.249817
0.5900	0.610256	0.610169	8.7×10^{-5}	0.457143
0.6100	0.721368	0.721311	5.6×10^{-5}	0.550916
0.6300	0.825397	0.825397	$< 10^{-6}$	0.785348
0.6500	0.923077	0.923077	$< 10^{-6}$	0.984615

Harmonic Kolmogorov–Smirnov statistic $D_{4095}^H = 2.44 \times 10^{-4}$.

Stern–Brocot empirical CDF $F_?$ shows large deviations from any smooth law.

The harmonic empirical CDF matches the theoretical $1/t^2$ density to four decimal places at every test point. The Stern–Brocot empirical CDF at the same test points shows large deviations: at $s = 0.55$ the harmonic CDF is at 0.36 while Stern–Brocot is at 0.09; at $s = 0.59$ the harmonic CDF is at 0.61 while Stern–Brocot is at 0.46. The Stern–Brocot empirical distribution is concentrating mass unevenly across the interval, with characteristic plateaux around prominent Stern–Brocot rationals such as $3/5$ and $5/8$. This non-uniformity is the signature of the singular Minkowski measure that the Stern–Brocot tree induces; the next subsection states the precise result.

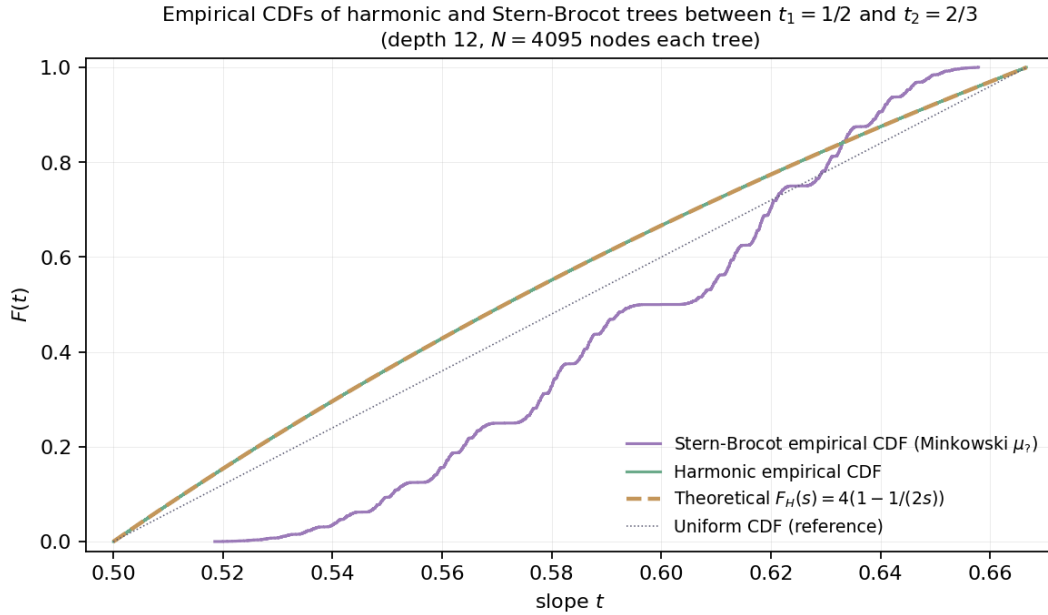


Figure 3: Empirical cumulative distribution functions of the harmonic and Stern–Brocot trees between seeds $t_1 = 1/2$ and $t_2 = 2/3$, depth 12, each tree containing $N = 4095$ nodes. The harmonic empirical CDF (green) lies exactly on the theoretical smooth curve $F_H(s) = 4(1 - 1/(2s))$ (amber, dashed). The Stern–Brocot empirical CDF (violet) shows the staircase structure of the Minkowski singular measure: long plateaux where no mass accumulates, separated by steep rises near prominent Stern–Brocot rationals like $3/5 = 0.6$ and $5/8 = 0.625$. The uniform CDF (grey, dotted) is included as a reference. See Appendix A.

The code in Appendix A illustrates that the empirical density of the harmonic tree is monotonically decreasing across the interval, as predicted by the $1/t^2$ form: the density ratio $\rho_H(t_1)/\rho_H(t_2) = (t_2/t_1)^2 = 16/9 \approx 1.78$, so the small-slope end of the interval is sampled about 78% more densely than the large-slope end. The Stern–Brocot tree, by contrast, visibly concentrates mass on a fractal subset of the interval rather than spreading it according to any smooth density at all.

5.3 The Stern–Brocot tree: singular Minkowski measure

The Stern–Brocot tree, traversed breadth-first, induces a measure of fundamentally different character.

Theorem 5.2 (Empirical measure of the Stern–Brocot tree). *The empirical distribution of the Stern–Brocot tree on $[0, 1]$, traversed breadth-first, converges to the Minkowski question-mark measure $\mu_?$: the probability measure whose cumulative distribution function is the Minkowski question-mark function $?(x)$. The measure $\mu_?$ is continuous and strictly increasing but singular with respect to Lebesgue measure: it has derivative zero almost everywhere, with all its mass concentrated on a set of Lebesgue measure zero.*

The Minkowski question-mark function was introduced by Minkowski in 1904 as a continuous bijection $[0, 1] \rightarrow [0, 1]$ mapping each rational to a dyadic rational and each quadratic irrational to a non-dyadic rational. The recursive definition descends the Stern–Brocot tree and maps each tree node at depth d to a dyadic rational of denominator 2^d . Salem (1943) proved that the function is singular: its derivative vanishes Lebesgue-almost-everywhere, yet the function rises continuously from 0 to 1 by accumulating its increase on the measure-zero set of irrationals.¹

	Stern–Brocot (mediant)	Harmonic
Subdivision rule	$(a + c)/(b + d)$	$2ac/(ad + bc)$
Depends on representatives	yes	no
Splitting ratio	depth-dependent integer arithmetic	$t_1 : t_2$
Projective character	coordinate-dependent	cross-ratio natural
Enumeration of $\mathbb{Q}$	complete	dyadic subset of $\mathbb{Q}$
Limit measure (BFS)	singular continuous	absolutely continuous
Density wrt Lebesgue	does not exist	$\rho_H(t) \propto 1/t^2$
Support	full interval topologically; measure-zero set measure-theoretically	full interval
CDF	Minkowski $?(x)$	$1 - \frac{t_1 t_2}{(t_2 - t_1)t} - \frac{t_1}{(t_2 - t_1)}$

The two measures are mutually singular: each places mass where the other places none. Both trees produce sequences dense in the seed interval. Both generate infinite families of primitive Pythagorean triples on the unit circle via the slope-to-triple map. They differ in the fine measure-theoretic structure of how they fill the interval.

¹The graph is described in the literature as a “slippery devil’s staircase”. The induced measure $\mu_?$, being the pushforward of uniform measure under $?^{-1}$, inherits the singular structure: it places no mass on any interval of positive Lebesgue measure but is supported on a Cantor-like fractal set. The references are classical and we do not reproduce the proof here; the standard treatments are Minkowski 1904, Denjoy 1938, Salem 1943, de Rham 1957.

6. The dynamical picture: both trees as doubling-map conjugates

The empirical measures of Section 5 can be derived without computation, as consequences of the ergodic theory of a single underlying dynamical system: the doubling map $D(u) = 2u \pmod{1}$ on $[0, 1]$. Each tree is conjugate to a canonical action whose invariant measure pushes forward through the conjugacy to give the empirical measure of the tree's breadth-first traversal. The harmonic and Stern–Brocot trees are then two faces of the doubling map seen through two different conjugacies.

6.1 The Stern–Brocot tree is not a linear action on slope pairs

It is natural to ask whether the harmonic recursion admits a $\mathrm{PSL}(2, \mathbb{Z})$ -style description. The Stern–Brocot tree does: writing a node as a 2×2 matrix $M = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ of boundary slopes $a/b, c/d$, the left and right children correspond to right-multiplication by $L = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ respectively. These generate $\mathrm{PSL}(2, \mathbb{Z})$, the modular group, and the Stern–Brocot tree is the orbit of the identity under their free monoid action.

Theorem 6.1 (Harmonic recursion is not given by linear action). *There is no fixed 2×2 matrix R_H such that right multiplication by R_H on the boundary matrix $M = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ produces the harmonic insertion $H(a/b, c/d) = 2ac/(ad + bc)$ as one of the new columns.*

Proof. The harmonic numerator $2ac$ is bilinear in a and c . Any product $M \cdot R_H$ has entries that are linear in each row of M when columns of R_H are fixed. A linear function of (a, c) cannot equal $2ac$ for all (a, c) . $\square$

The harmonic tree therefore does not live in $\mathrm{PSL}(2, \mathbb{Z})$ or in any matrix semigroup acting by ordinary linear Möbius transformations on the boundary pair. To find its natural dynamics we change coordinates.

6.2 The harmonic tree is the doubling map in reciprocal coordinates

Set $u = 1/t$. The harmonic mean becomes the arithmetic mean, $1/H(t_1, t_2) = \frac{1}{2}(1/t_1 + 1/t_2) = A(u_1, u_2)$. Under this coordinate change the harmonic recursion is exact dyadic bisection of the interval $[u_2, u_1]$, where $u_i = 1/t_i$ for seeds $t_1 < t_2$.

Theorem 6.2 (Harmonic tree = doubling map). *Let $\phi : [t_1, t_2] \rightarrow [0, 1]$ be the affine reparametrisation in reciprocal coordinates:*

$$\phi(t) = \frac{1/t_1 - 1/t}{1/t_1 - 1/t_2}.$$

Then ϕ conjugates the breadth-first traversal of the harmonic tree to dyadic enumeration of $[0, 1]$, and conjugates the harmonic empirical measure to Lebesgue measure on $[0, 1]$. The doubling map $D(x) = 2x \pmod{1}$ preserves this Lebesgue measure and is ergodic with respect to it.

Proof. The harmonic insertion of $H(t_1, t_2)$ between seeds t_1, t_2 satisfies $1/H = \frac{1}{2}(1/t_1 + 1/t_2)$, so $\phi(H(t_1, t_2)) = \frac{1}{2}$, the midpoint of $[0, 1]$. The recursive insertions at all depths map to dyadic rationals $k/2^d$ for $1 \leq k \leq 2^d - 1$. Breadth-first enumeration of the harmonic tree therefore corresponds to depth-by-depth enumeration of dyadic rationals, which is equidistributed with respect to Lebesgue measure. The doubling map D acts on the dyadic

rationality by left bit-shift in binary expansion, with Lebesgue measure as its unique absolutely continuous invariant measure, ergodic by a standard argument. $\square$

The harmonic limit density on $[t_1, t_2]$ is then the pushforward of Lebesgue measure on $[0, 1]$ under ϕ^{-1} . Computing: $\phi^{-1}(x) = t_1 t_2 / (t_2 - (t_2 - t_1)x)$, and $|d\phi/dt| = t_1 t_2 / ((t_2 - t_1)t^2)$, so the pushforward density on t is exactly

$$\rho_H(t) = \frac{t_1 t_2}{(t_2 - t_1)t^2},$$

recovering Theorem 5.1 as a one-line consequence of the conjugacy.

6.3 The Stern–Brocot tree is the doubling map under the Minkowski conjugacy

The Stern–Brocot tree’s natural dynamics is the *Gauss map*

$$G(x) = \left\{ \frac{1}{x} \right\}, \quad x \in (0, 1),$$

the fractional part of the reciprocal. The Gauss map governs continued fractions: iterating G on x recovers the continued-fraction digits of x , and walking the Stern–Brocot tree to approximate x recovers the convergents of those digits. The Gauss map preserves the *Gauss measure* $d\mu_G = dx / ((1+x) \log 2)$, which is absolutely continuous and ergodic.

The connection to the doubling map is the *Minkowski question-mark function* $?(x)$, defined by mapping each Stern–Brocot rational at depth d to the dyadic rational of denominator 2^d at the corresponding position. The classical identity (Denjoy 1938, Salem 1943) is

$$?(G(x)) = D(?(x)) = \{2 \cdot ?(x)\}, \quad (7)$$

expressing that $?$ conjugates the Gauss map G to the doubling map D .

Theorem 6.3 (Stern–Brocot tree = doubling map via $?$). *The Stern–Brocot empirical measure on $[0, 1]$ (breadth-first traversal) is the pushforward of Lebesgue measure on $[0, 1]$ under the inverse Minkowski function $^{-1}$. Equivalently, the question-mark function $?$ is the cumulative distribution function of the Stern–Brocot empirical measure. Since $?$ is continuous and strictly increasing but has derivative zero almost everywhere (Salem 1943), the pushforward measure $\mu_? = (^{-1})_* \text{Leb}$ is singular continuous with respect to Lebesgue.*

This is essentially the definition of the question-mark function, recast as a statement about the Stern–Brocot tree’s empirical distribution. The singularity of $\mu_?$, asserted in Section 5, is now a consequence of Salem’s classical result on the regularity of $?$ rather than an empirical observation.

6.4 Both trees, one underlying dynamical system

The dynamical-systems picture aligns the two trees as conjugates of the same map:

	Harmonic tree	Stern–Brocot tree
Conjugating map	$\phi(t) = \text{affine in } 1/t$	Minkowski $?(x)$
Conjugacy regularity	analytic	singular continuous
Conjugate map	doubling $D(x) = 2x \bmod 1$	doubling $D(x) = 2x \bmod 1$
Underlying measure on $[0, 1]$	Lebesgue dx	Lebesgue dx
Pushforward to t -line	$\rho_H(t) \propto 1/t^2$ (smooth)	$\mu_?$ (singular)

Both trees are conjugates of the doubling map on $[0, 1]$, equipped with its Lebesgue invariant measure. The harmonic conjugating map is analytic (the reciprocal-affine map ϕ), so the pushforward is absolutely continuous and the density is elementary. The Stern–Brocot conjugating map is the Minkowski $?$ function, which is singular, so the pushforward inherits singularity. The empirical-measure results of Section 5 are not coincidences: they are the two natural ways to coordinatise the doubling map’s invariant Lebesgue measure on the slope axis, one through a smooth diffeomorphism and one through the question-mark function.

Remark 6.4 (Ergodicity). Since the doubling map is ergodic with respect to Lebesgue measure on $[0, 1]$, the Birkhoff ergodic theorem applies: for any integrable f on $[0, 1]$,

$$\frac{1}{N} \sum_{k=0}^{N-1} f(D^k(x)) \longrightarrow \int_0^1 f(y) dy \quad \text{for Lebesgue-a.e. } x.$$

Through the conjugacies, the same ergodic theorem applies to observables along the two trees: time averages of any continuous function along the breadth-first sequence of either tree converge to the spatial integral against the appropriate pushforward measure ($\rho_H dt$ for harmonic, $d\mu_?$ for Stern–Brocot). The word “ergodic” is here used in its precise dynamical sense, not merely as a synonym for “equidistribution”: both trees inherit the ergodicity of the doubling map through their respective conjugacies.

7. The Bertrand question, answered by recursion

The Bertrand framing of the opening section now bears the weight of the analysis. Restricting the original paradox to chords through a fixed boundary point reduces the chord space to a one-dimensional pencil, parametrised by the slope axis $\mathbb{P}^1(\mathbb{R})$. On this pencil collapses the classical family of Euclidean-invariant parametrisations to a single canonical rotationally invariant measure: it is the uniform angular measure $d\mu_\theta$ on the circle, equivalently the Cauchy measure $dt/(1+t^2)$ in slope coordinates. The classical multiplicity of Bertrand sampling rules does not survive the restriction; only this one natural Euclidean measure remains on the pencil.

Three measures on the pencil. The pencil therefore admits three natural probability measures of distinctly different character, all living on the same one dimensional space:

1. The *Euclidean uniform angular measure*: $d\mu_\theta \propto d\theta = 2 dt/(1+t^2)$. Absolutely continuous, smooth, derived from continuous geometric sampling of the second endpoint on the circumference. This is the only natural Euclidean measure on the restricted chord pencil.
2. The *Stern–Brocot recursive measure*: the Minkowski question-mark measure $\mu_?$, induced by breadth-first traversal of the mediant subdivision $(a/b, c/d) \mapsto (a+c)/(b+d)$. Continuous and strictly increasing, but singular with respect to Lebesgue measure; derivative zero almost everywhere.
3. The *harmonic recursive measure*: absolutely continuous with density $\rho_H(t) \propto 1/t^2$ on the seed interval, induced by breadth-first traversal of the harmonic subdivision $(a/b, c/d) \mapsto 2ac/(ad+bc)$.

The first measure is Euclidean and continuous. The other two are not Euclidean at all: they

arise from purely arithmetic recursive subdivision of the slope axis. The mediant, being coordinate-dependent on integer representatives and arithmetically complete, induces a fractal singular measure. The harmonic mean, being projectively natural via the cross-ratio characterisation of Theorem 3.1 and arithmetically sparse (a dyadic subset of the rationals under the reciprocal-coordinate identification), induces a smooth absolutely continuous measure with elementary explicit density. The three measures are mutually singular as pairs except where one of them is Lebesgue-equivalent; the Euclidean and harmonic measures share absolute continuity but have different densities, while the Stern–Brocot measure is singular with respect to both.

The Bertrand paradox does not directly extend to the fixed-basepoint pencil, because the multiplicity of natural Euclidean measures is killed by the dimensional reduction. What does extend is the underlying lesson: a one-dimensional space, even one with a distinguished Euclidean measure, can support further natural probability measures from sources outside the Euclidean tradition. Recursive arithmetic subdivision is one such source, and the choice of subdivision rule determines a measure whose regularity class is a structural property of the recursion, not of the Euclidean geometry.

The Bertrand-style question, restated for recursive measures on the slope axis, becomes:

Choose a random rational slope from $(-1, 0)$ by recursive arithmetic subdivision. What is the probability that the resulting chord is longer than the side of the inscribed equilateral triangle? The answer depends on which subdivision rule generates the slope, and the two natural choices give answers in different regularity classes. Under the harmonic recursive measure the answer is computable in closed form from the smooth $1/t^2$ density; under the Stern–Brocot recursive measure the answer is a value of the Minkowski question-mark function, computable but with no elementary closed form.

Geometry alone does not determine probability. Continuous Euclidean sampling does not exhaust the natural measures. Recursive subdivision is itself a source of probability measures on the projective line, and the choice of subdivision rule is the choice of measure. This is the contribution of the present paper to the broader program initiated in *From Bertrand to Farey*: a recursive measure distinct in regularity from both the Euclidean uniform measure and the Minkowski measure, generated by the projectively natural harmonic interpolation.

Closing Remarks

The harmonic interpolation tree suggests a probability measure induced by recursive subdivision itself. The recursion determines how mass is distributed across the projective line, and different subdivision rules generate different invariant measures. The Farey mediant produces the singular Minkowski measure. The harmonic interpolation produces the absolutely continuous density

$$d\mu_H(t) \propto \frac{dt}{t^2}.$$

Thus the contribution of this paper is the observation that recursive arithmetic processes naturally generate probability measures on projective space. In this view, recursion itself becomes a mechanism for assigning probability, placing arithmetic subdivision alongside the classical geometric constructions as a distinct source of measure. Geometry determines the space; recursion determines the measure.

A. Python code of CDF

The numerical verification of Theorem 5.1 and the comparison figure in Section 5.2 were produced by the following Python code. The script builds both the harmonic and Stern–Brocot trees breadth-first to depth 12 between seeds $t_1 = 1/2$ and $t_2 = 2/3$, computes the empirical cumulative distribution function of each, compares the harmonic empirical CDF with the theoretical $F_H(s) = (t_2/(t_2 - t_1)) \cdot (1 - t_1/s)$, and plots all three (both empirical CDFs and the theoretical curve) on the same axes for visual comparison.

```
from fractions import Fraction
import matplotlib.pyplot as plt
import numpy as np

# Insertion rules

def H(a, b):
    """Harmonic mean of two positive Fractions: 2ab/(a+b)."""
    return Fraction(
        2 * a.numerator * b.numerator,
        a.numerator * b.denominator + b.numerator * a.denominator,
    )

def mediant(a, b):
    """Farey mediant: (p1+p2)/(q1+q2)."""
    return Fraction(a.numerator + b.numerator, a.denominator + b.denominator)

def tree_breadth(t1, t2, rule, max_depth):
    """All nodes of the binary subdivision tree between t1 and t2,
    breadth-first, up to max_depth, using the given insertion rule."""
    nodes = []
    boundaries = [(t1, t2)]
    for _ in range(1, max_depth + 1):
        new_bounds = []
        for a, b in boundaries:
            m = rule(a, b)
            nodes.append(m)
            new_bounds.append((a, m))
            new_bounds.append((m, b))
        boundaries = new_bounds
    return nodes

# Build both trees to depth 12

t1, t2 = Fraction(1, 2), Fraction(2, 3)
DEPTH = 12

harm_nodes = tree_breadth(t1, t2, H, DEPTH)
mdnt_nodes = tree_breadth(t1, t2, mediant, DEPTH)

ts_harm = sorted(float(x) for x in harm_nodes)
ts_mdnt = sorted(float(x) for x in mdnt_nodes)
N = len(ts_harm) # 4095

# Theoretical harmonic CDF

def F_harm_theory(s):
    a, b = float(t1), float(t2)
    return (b / (b - a)) * (1 - a / s)
```

```

# Pointwise comparison table

for s in [0.51, 0.53, 0.55, 0.57, 0.59, 0.61, 0.63, 0.65]:
    F_h_emp = sum(1 for x in ts_harm if x <= s) / N
    F_h_th = F_harm_theory(s)
    F_m_emp = sum(1 for x in ts_mdnt if x <= s) / N
    print(f"s={s:.4f}  F_H_emp={F_h_emp:.6f}  "
          f"F_H_th={F_h_th:.6f}  F_med={F_m_emp:.6f}")

D_harm = max(abs((i+1)/N - F_harm_theory(t)) for i, t in enumerate(ts_harm))
print(f"Harmonic KS statistic D_N = {D_harm:.6f}  (N = {N})")

# Plot all three CDFs together

fig, ax = plt.subplots(figsize=(7.5, 4.6), dpi=120)

F_emp = np.arange(1, N + 1) / N
ax.step(ts_mdnt, F_emp, where="post", color="#9b7ab8",
        linewidth=1.4, label="Stern-Brocot empirical CDF (Minkowski)")
ax.step(ts_harm, F_emp, where="post", color="#6baa8a",
        linewidth=1.4, label="Harmonic empirical CDF")

s_grid = np.linspace(float(t1) + 1e-6, float(t2), 400)
ax.plot(s_grid, [F_harm_theory(s) for s in s_grid],
        color="#c4965a", linewidth=2.0, linestyle="--",
        label="Theoretical $F_H(s) = 4(1 - 1/(2s))$")

ax.plot([float(t1), float(t2)], [0, 1], color="#5a5a72",
        linewidth=0.7, linestyle=":", label="Uniform CDF (reference)")

ax.set_xlabel("slope $t$")
ax.set_ylabel("$F(t)$")
ax.legend(loc="lower right", fontsize=8.5, frameon=False)
ax.grid(True, alpha=0.25, linewidth=0.4)
plt.tight_layout()
plt.savefig("cdf_comparison.pdf", bbox_inches="tight")

```

Running this code produces Figure 3 and the verification table in Section 5.2. The KS statistic for the harmonic tree, $D_N \approx 2.44 \times 10^{-4}$ at $N = 4095$, is well within the expected $O(1/\sqrt{N})$ convergence bound and confirms the theoretical density. For the Stern–Brocot tree, no comparable KS test to a smooth limit is meaningful, because the limit is singular: the empirical CDF is approximating a function whose derivative is zero almost everywhere, not a function with any pointwise density.